Do meritocratic beliefs undermine the acceptance of privilege at school-age? A longitudinal equivalence approach

Abstract

Schools are key spaces of meritocratic socialization, yet little is known about how students conceive meritocracy and how these conceptions relate to the acceptance of privilege. Using an instrument previously applied to adults, we surveyed Chilean students in 6th and 9th grade (Wave 1: N = 846) and again in 7th and 10th grade (Wave 2: N = 662). Results support four distinct dimensions of beliefs, equivalent across waves but only partially across school stages. Endorsing meritocracy did not imply rejecting privilege: about one third of students held both positions, although their latent association was weak and inconsistent across groups. Latent means did not change significantly over one year. Secondary students were more skeptical than primary students about whether effort is rewarded, a cross-sectional contrast limited by item non-equivalence. The study offers a conceptual and measurement foundation for research on the development of these beliefs and their educational implications.

Keywords: Meritocracy · Privilege · Schools · Measurement invariance · Socialization · Chile

Word count: 6658 (main text, notes, statements and references; abstract, tables and figures excluded)

1 Introduction

The way in which children and youths come to understand who succeeds in society and why is a relevant aspect of the socialization process (Mijs, 2016; Resh & Sabbagh, 2014). Schools are important actors in this regard, as they use grading, tracking, and selection to apply meritocratic criteria based primarily on individual talent and effort (Darnon et al., 2018; Traini, 2022). This early exposure to meritocratic principles in the "impressionable years" (Krosnick & Alwin, 1989) shapes students' understanding of fairness, social mobility, and social hierarchies. At the same time, schools are environments where students encounter evidence that outcomes are shaped by factors other than meritocratic ones, such as family background, social connections, and structural advantages usually associated with privilege (Bourdieu & Passeron, 1990; Goudeau & Croizet, 2017; Tang et al., 2025). These background attributes can be considered privileges due to heritage-based advantages over which people have no control, unlike effort. Yet, few studies have explored how students conceive meritocracy and to what extent such conceptions challenge the role of privilege in society.

Although meritocracy is often treated as a single dimension, mostly related to individual effort, the literature increasingly suggests that it is a multidimensional construct. On the one hand, this refers to the distinction between perceptions of how meritocracy operates in society ("what is") and preferences about how it should operate ("what ought to be") (Castillo et al., 2023; Duru-Bellat & Tenret, 2012). On the other hand, meritocratic beliefs can be distinguished from privilege-based beliefs, which emphasize the role of family wealth and social connections in shaping rewards in society (C. Liu & Wang, 2025; Tang et al., 2025). A multidimensional approach should therefore incorporate perceptions and preferences about both meritocratic and privilege-based explanations of social outcomes, in line with Khan's (2013) call to analyze meritocracy together with privilege.

Previous research has shown that privilege and meritocratic beliefs represent different and not necessarily opposite dimensions, therefore they can coexist in people's minds (Fan et al., 2026; C. Liu & Wang, 2025; Tang et al., 2025). However, attempts to study these beliefs have been limited by the lack of instruments that capture their multidimensionality and by the absence of longitudinal studies that examine how these beliefs evolve over time. We contribute to this agenda by applying a multidimensional model to measure meritocratic and privilege beliefs in a sample of Chilean students and by examining the stability and measurement equivalence of these beliefs across school stages and over time. We argue that meritocratic and privilege beliefs can be distinguished even at an early school age, with a structure similar to that found among adults (Castillo et al., 2023); that they are not two ends of the same continuum but can coexist in students' minds; and that they are not static but evolve as students accumulate experience and exposure to different social contexts.

We draw on a two-wave panel of Chilean students in primary and secondary school cohorts. Chile's educational system combines high stratification, market-based competition, and explicit meritocratic discourse in school culture (Castillo et al., 2024; Valenzuela et al., 2013), alongside persistent socioeconomic inequality and a deep marketization of social welfare (Ferre, 2023; Madariaga, 2020). This makes Chile a productive case for studying how students internalize, question, or reconcile meritocratic ideals with the reality of privilege.

The research questions guiding this study are as follows:

(1) Are school-aged students able to distinguish between meritocratic and privilege-based principles, as well as between perceptions ("what is") and preferences ("what should be")?

(2) Are meritocratic beliefs undermined (i.e., negatively associated) by beliefs about privilege, or could they coexist in students' minds?

(3) Are these dimensions stable over time and across school stages?

2 Theoretical and empirical background

2.1 The socialization of meritocracy at school

Meritocracy refers to a distributive system in which individual merit, typically defined as effort and personal ability, is treated as the primary criterion for allocating resources and rewards, rather than social origins or inherited privilege (Young, 1958). While Young (1958) coined the term as a dystopian critique, it has since been re-appropriated as a positive ideal of fairness in liberal and market-oriented societies (Mijs, 2021; Van De Werfhorst, 2024). Because it frames outcomes as earned, meritocracy can reinforce inequality: "winners" see their position as deserved, while "losers" are pushed toward self-blame rather than structural critique (García-Sierra, 2023; Sandel, 2020). In this line, research in adult populations has documented systematic associations between meritocratic beliefs and the justification of social inequalities (C. Liu & Wang, 2025; Mijs, 2021; Tejero-Peregrina et al., 2025). Yet how and when these beliefs develop, and whether they can be identified at early stages of socialization, has received little attention.

Schools are key institutions when it comes to the socialization of meritocracy. On the one hand, they advance meritocratic ideals through narratives of effort, achievement, self-improvement, and education as a legitimate route to rewards and social mobility (Batruch et al., 2022; Chauvin et al., 2026; Darnon et al., 2018; Wiederkehr et al., 2015). On the other hand, they embed these ideals in routine practices of grading, tracking, selection, awards, and credentialing, which make merit visible and consequential (Lampert, 2013; Resh & Sabbagh, 2014; Traini, 2022). At the same time, schools are also settings where privilege-based advantages are visible and consequential, such as family resources, cultural capital, peer environments, and heterogeneous school quality (Batruch et al., 2022; Bernstein, 1996; Goudeau & Croizet, 2017; Khan & Jerolmack, 2013; Lareau, 2003). Students may endorse meritocracy as a normative ideal while simultaneously recognizing structural constraints, reflecting what the literature describes as dual consciousness (Kluegel & Smith, 1987; C. Liu & Wang, 2025; Tang et al., 2025).

The dynamics of the socialization of meritocracy can change over time. Developmental research suggests that by around age 10, children already articulate judgments about social differences, typically emphasizing individual, merit-based explanations while only beginning to incorporate opportunity-based accounts (Imhoff & Brussino, 2025; Sigelman, 2013). As students move through the school system, however, cumulative exposure to grading, tracking, peer comparison, and visible socioeconomic gaps can make the limits of effort as a sufficient explanation for success increasingly apparent. Tang et al. (2025) show that among Chinese college students, those from lower socioeconomic backgrounds initially report lower recognition of privilege-based advantages than their higher-SES peers, but this gap narrows progressively across college years as both groups converge toward stronger acknowledgment of structural advantage. This suggests that sustained

exposure to the educational system makes structural advantage harder to ignore–referred to as a "reality shock" (Allen, 2016). Adolescence is therefore a particularly consequential period for studying these processes, precisely when abstract reasoning about justice and fairness becomes more sophisticated and school-based sorting mechanisms grow more salient (Henry & Saul, 2006; Resh & Sabbagh, 2014).

2.2 Conceptualizing and measuring meritocratic and privilege beliefs

Most attempts to operationalize meritocratic beliefs often follow Young's (1958) formulation (Merit = Effort + Intelligence). For instance, the "Preference for the Merit Principle Scale" (Davey et al., 1999) and the "Belief in School Meritocracy Scale" (Batruch et al., 2022) focus on the perceived importance of effort and ability for success, while Wiederkehr et al. (2015) measure the extent to which respondents believe that hard work is rewarded. These approaches, however, risk conflating meritocratic beliefs with general attitudes toward effort or achievement, potentially overlooking the broader structural and relational dimensions of meritocracy.

A further limitation in the conceptualization of meritocratic beliefs concerns privilege-based factors such as family wealth, social connections, and other structural advantages (Khan & Jerolmack, 2013). As Castillo et al. (2023) notes, much survey research relies on item batteries that ask respondents to rate the importance of effort, talent, family background, networks, or luck for "getting ahead", and then reduces these into a single score. This becomes especially problematic when researchers construct continuum measures in composite indicators, treating meritocratic and privilege-based explanations as opposite ends of the same scale (e.g., by subtracting "non-merit" from "merit," as in Reynolds & Xian, 2014). Such operationalizations embed a strong zero-sum assumption, namely that endorsing merit necessarily entails rejecting structural or relational advantages, thereby ruling out, by design, the empirical possibility that both types of beliefs coexist. Recent debates, including the exchange between Mijs (2026) and Wiesner & Sachweh (2026), underscore this point by arguing for measures that capture meritocratic beliefs alongside privilege-based ones, rather than as a residual of the relative importance assigned to structural factors Empirical work points in the same direction: meritocratic endorsement may remain stable as recognition of structural constraints increases (C. Liu & Wang, 2025; Tang et al., 2025), support for merit can coexist with recognition of social connections in distinct attitudinal clusters (Kwon & Pandian, 2024), and meritocratic and structural explanations often accumulate rather than replace one another (Zhu, 2025). Unidimensional indices, and especially subtraction scores, can therefore manufacture artificial neutral positions and blur substantively distinct belief profiles that could hide relevant group differences and changes over time.

In response to the previously mentioned limitations, Castillo et al. (2023) proposed a multidimensional framework that decomposes meritocratic beliefs along two analyti-

cally independent axes: perceptions (what is) versus preferences (what ought to be), and meritocratic versus privilege-based (originally called "non-meritocratic") allocation principles. Meritocratic elements (effort, talent) and privilege-based elements (family wealth, networks) are treated as distinct components rather than as opposites. This yields four separate constructs, allowing belief combinations that older measures tend to collapse, such as endorsing meritocracy as an ideal while simultaneously recognizing the weight of family background and connections (Khan & Jerolmack, 2013). Evidence from adult samples using confirmatory factor analysis supports a four-factor structure (perceptions and preferences of both meritocracy and privilege) (Castillo et al., 2023).

Attention to measurement quality extends beyond technical considerations and constitutes a matter of sociological concern. As the sociological discussion on estimands Lundberg et al. (2021) stresses, constructs must be theoretically meaningful and empirically identifiable; otherwise, conclusions may reflect artifacts of measurement rather than substantive social phenomena. Invariance analysis assesses whether the same latent construct is measured across groups or over time, a precondition for valid comparisons and developmental inference (Davidov et al., 2014; Leitgöb et al., 2023; Putnick & Bornstein, 2016). Beyond establishing comparability, it identifies which aspects of a construct are equivalent and which are not, which is particularly relevant for students' meritocratic and privilege beliefs, whose socialization may change over time and acrossschool stages (C. Liu & Wang, 2025; Tang et al., 2025).

Building on the previous background, we state the following hypotheses:

H_1 **(Dimensional differentiation)**: Students' meritocratic and privilege beliefs can be represented and measured by four distinct dimensions: perceived meritocracy, perceived privilege, meritocratic preferences, and privilege acceptance.

H_2 **(Beliefs compatibility)**: merit and privilege beliefs are not mutually exclusive, but rather compatible orientations that can coexist within the same individual. This means that endorsing merit does not necessarily imply rejection of privilege.

H_3 **(Equivalence)**: The four-factor model is equivalent across the two waves of the panel survey and across two student cohorts – 6th and 9th grade (or first-year secondary) in the first wave.

3 Methods

3.1 Participants

This study uses secondary data from the 2023 and 2024 waves of a panel survey on education and meritocracy [name withheld for anonymous review]. The panel examines school-age students' beliefs, attitudes, and behaviors regarding meritocracy, inequality, and citizenship.

The questionnaire was administered to sixth-grade and ninth-grade students (primary

and secondary cohorts, respectively) in nine schools in Chile's Metropolitan and Valparaíso regions, a smaller number than the 15 schools originally planned, owing to refusals from selected schools. Although the sample was non-probabilistic and lacked quota controls, a minimum target of 900 students was set to ensure adequate statistical power. After data processing and listwise deletion, the analytical sample included 846 students in Wave 1 and 662 in Wave 2 (see Table 1 for age and gender by cohort). Of the 846 Wave 1 students, 591 also had complete data in Wave 2 (linked retention = 69.9%), and the Wave 2 sample includes 71 students without complete Wave 1 data. The main sources of attrition were school transfers between academic years and high absenteeism in several schools. Within cohorts, the Wave 1 item means of retained and non-retained students differed by at most 0.14 points on the four-point scale.

Table 1: Descriptive statistics for cohort level

			Age		Gender		
Wave	level	N	Mean	SD	Man	Women	Others
Wave 1	Primary	406	11.85	0.56	195	191	20
	Secondary	440	14.79	0.80	226	195	19
	Total	846	13.38	1.63	421	386	39
Wave 2	Primary	319	12.83	0.55	161	145	13
	Secondary	343	15.76	0.75	177	158	8
	Total	662	14.35	1.61	338	303	21

3.2 Procedure

Data were collected by a professional research firm using a Computer-Assisted Web Interviewing (CAWI) approach, based on online questionnaires. Written informed consent was obtained for all participants: students signed the consent form, which was reviewed and approved by a parent or legal guardian, and only students with signed consent were included in the study. To enable longitudinal linkage while protecting confidentiality, the survey firm generated anonymous unique identifiers for each student, and no directly identifying information was included in the analytic files.

The study was approved by the Research Ethics Committee of [institution withheld for anonymous review] (reference number withheld for anonymous review), which determined that the project posed minimal risk and approved the consent and confidentiality procedures. The hypotheses were preregistered on the Open Science Framework (OSF) prior to data analysis [link withheld for anonymous review].

3.3 Measures

3.3.1 Perceptions and preferences of meritocracy and privilege

Meritocratic and privilege-based perceptions and preferences were measured with the eight items of the original scale (Castillo et al., 2023), two per dimension :

- Meritocratic perceptions: the extent to which effort and ability are rewarded in Chile,
- Privilege perceptions: the extent to which success is perceived as linked to connections and family wealth,
- Meritocratic preferences: agreement that those who work harder or are more talented should be better rewarded,
- Privilege acceptance: agreement that it is acceptable for individuals with better connections or wealthy parents to achieve greater success.

Each item (see Table 2) is rated on a four-point scale ranging from "strongly disagree" (1) to "strongly agree" (4); higher scores indicate stronger endorsement of the corresponding perception or preference. The original Spanish wording of the items is reported in the supplementary material.

Table 2: Items for perceptions and preferences for meritocracy scale

Components	Dimensions	Item
Perception	Meritocratic	In Chile people are rewarded for their efforts
		In Chile people are rewarded for their intelligence and ability
	Privilege	In Chile those with wealthy parents do much better in life
		In Chile those with good contacts do much better in life
Preference	Meritocratic	Those who work harder should reap greater rewards than those who work less hard
		Those with more talent should reap greater rewards than those with less talent
	Privilege	It is good that those who have rich parents do better in life
		It is good that those who have good contacts do better in life

3.4 Analytical strategy

3.4.1 Measurement model and estimation

We estimated a Confirmatory Factor Analysis (CFA) with four latent factors, using Diagonally Weighted Least Squares with robust correction (WLSMV), which is suitable for ordinal

items (Kline, 2023). Model fit was assessed following the guidelines of Brown (2015), using several indices: the Comparative Fit Index (CFI) and the Tucker-Lewis Index (TLI), with acceptable values above 0.95; the Root Mean Square Error of Approximation (RMSEA), with values below 0.06 indicating good fit; and the Chi-square statistic (acceptable fit indicated by $p > 0.05$ and a Chi-square/df ratio < 3).

3.4.2 Measurement equivalence across time and groups

A key contribution of this study is its assessment of measurement stability through factorial invariance testing (Putnick & Bornstein, 2016), since only if the same latent construct is measured in the same way across time and groups can observed differences be attributed to true differences rather than measurement artifacts. We tested invariance across the two panel waves (longitudinal invariance) and across school levels (primary vs. secondary cohorts), comparing nested models with increasing constraints: configural (same factor structure), weak (equal factor loadings), strong (also equal thresholds/intercepts), and strict (also equal residual variances).

We followed recent recommendations for ordered categorical indicators, specifically the longitudinal approach of Y. Liu et al. (2017) and the cross-group procedure as suggested by Svetina et al. (2020). In addition to the chi-square difference test, we assessed invariance using changes in approximate fit indices. Following Chen (2007), we treated ΔCFI as the primary criterion and ΔRMSEA as supplementary, indicating noninvariance when $\Delta\text{CFI} \leq -0.010$ and $\Delta\text{RMSEA} \geq 0.015$.

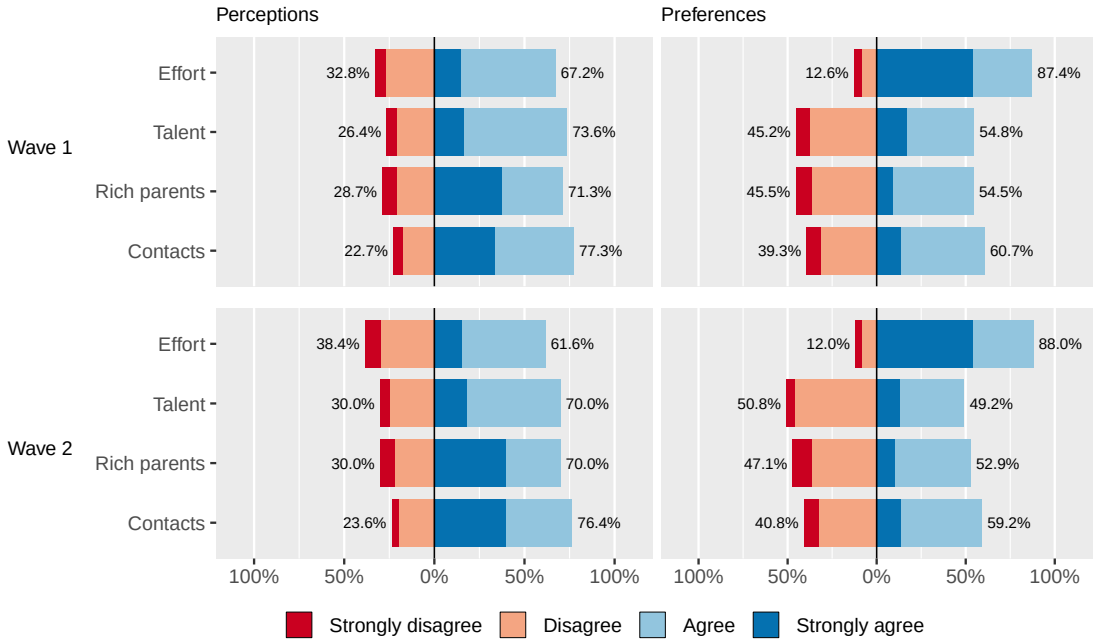
All analyses were performed using the *lavaan* package (Rosseel, 2012) in R version 4.5.2.

4 Results

4.1 Descriptives

Figure 1 displays the response distributions by wave, distinguishing perceptions from preferences. The pattern is stable across waves. Among perceptions, agreement is high for contacts, parental wealth, and talent, and lowest for effort. Among preferences, by contrast, effort is overwhelmingly endorsed as the legitimate basis for rewards, far exceeding its perceived role in practice, whereas talent is the least endorsed criterion, below contacts and family resources. More than half of respondents in both waves accept the influence of rich parents and contacts, so privilege criteria are not uniformly seen as an illegitimate way to get ahead.

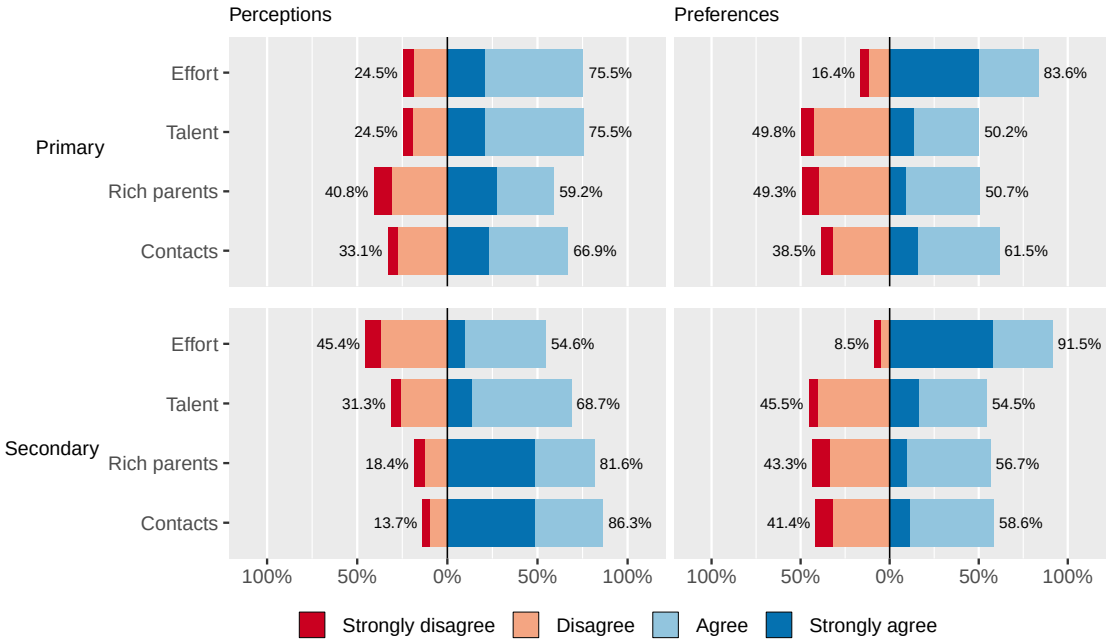
Figure 1: Distribution of responses on the scale of perceptions and preferences regarding meritocracy by wave



Source: own elaboration based on panel survey data [name withheld for review] (N Wave 1 = 846; N Wave 2 = 662)

Figure 2 shows the same items by cohort, which reveals a sharper contrast between perceptions and preferences. Meritocratic perceptions (effort and talent) receive more agreement in the primary than in the secondary cohort, while the reverse holds for privilege perceptions (contacts and family wealth), consistent with a "disillusionment" with meritocracy as students progress through school (Fan et al., 2026; Tang et al., 2025). Preferences are more stable across cohorts: effort is the most endorsed basis for reward, while talent, family wealth, and especially contacts remain more ambivalent, with contacts more often accepted than rejected in both cohorts. Overall, the cohort comparison reveals a widening gap between an effort-centered meritocratic ideal and the perceived importance of privilege in practice.

Figure 2: Distribution of responses on the scale of perceptions and preferences regarding meritocracy by cohort

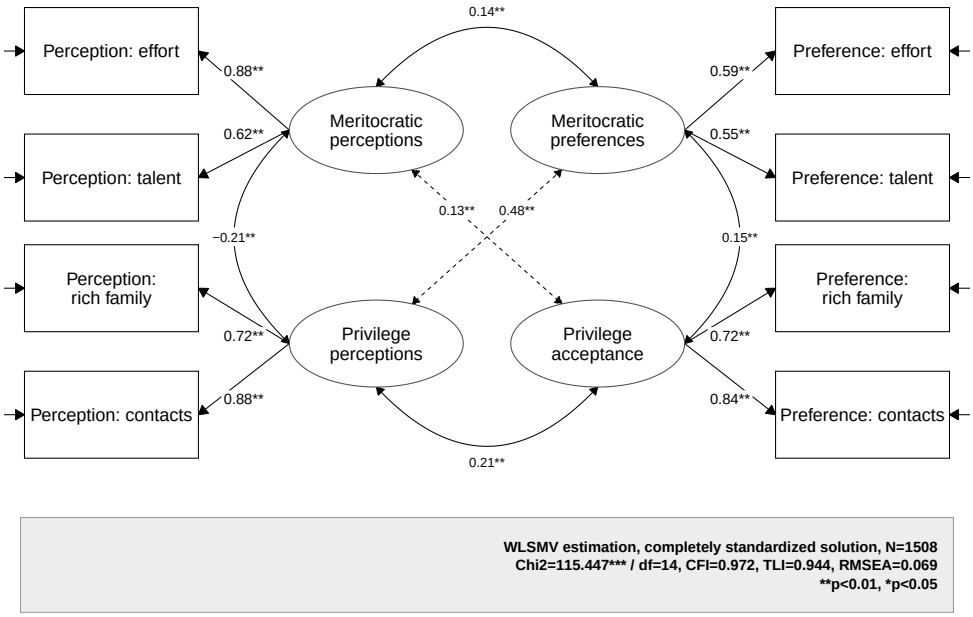


Source: own elaboration based on panel survey data [name withheld for review] (N Primary = 725; N Secondary = 783)

4.2 Multidimensional measurement of meritocracy and privilege

Figure 3 presents the measurement model for the pooled sample (both waves and cohorts), with the four latent dimensions (circles) each measured by two observed indicators (rectangles). Fit is reasonable but not fully satisfactory (CFI = 0.97; TLI = 0.94; RMSEA = 0.07): the CFI falls within commonly accepted ranges, but the TLI and RMSEA do not meet conventional thresholds. Part of this misfit relates to differences across cohorts, examined in the multigroup analyses below.

Figure 3: Measurement model diagram (pooled sample)



All standardized factor loadings are statistically significant and moderate to strong: effort (β = 0.88) and talent (β = 0.62) for meritocratic perceptions; effort (β = 0.59) and talent (β = 0.55) for meritocratic preferences, the weakest pair; contacts (β = 0.88) and rich family (β = 0.72) for perceived privilege; and contacts (β = 0.84) and rich family (β = 0.72) for privilege acceptance. Overall, the indicators are reasonably good measures of their respective latent constructs.

The latent correlations address H_2. The dimensions are interrelated but far from perfectly aligned, and all correlations but one are positive. The only negative correlation is between meritocratic and privilege perceptions (r = −0.21): perceiving society as more meritocratic goes with a lower recognition of the role of privilege. As among adults (Castillo et al., 2023), meritocratic perceptions and preferences are positively correlated (r = 0.14). The strongest association is between privilege perceptions and meritocratic preferences (r = 0.48): students who recognize the role of privilege in society are more likely to endorse meritocratic ideals. Less intuitively, privilege acceptance is positively correlated with both meritocratic perceptions (r = 0.13) and meritocratic preferences (r = 0.15). Although modest, these last correlations indicate that merit and privilege beliefs are not simply opposites.

4.3 Stability of meritocracy and privilege beliefs

4.3.1 Longitudinal invariance

Table 3 reports the longitudinal invariance tests, in which the same students measured at two time points are treated as groups and the unique variances of each item are corre-

lated across waves (Widaman et al., 2010). The scale achieved the strongest level, strict invariance: the strict model fit well, $\chi^2(84) = 130.17$, CFI = 0.991, RMSEA = 0.031 (90% CI: 0.020–0.040), and its fit was practically identical to that of the strong model, $\Delta\chi^2(4) = 1.635$, ΔCFI = 0.000, ΔRMSEA = −0.002, well within commonly used cutoffs (Chen, 2007).

Table 3: Longitudinal invariance results

Model	χ^2 (df)	CFI	RMSEA (90% CI)	$\Delta\chi^2$ (Δdf)	ΔCFI	ΔRMSEA	Decision
Configural	117.7 (68)	0.991	0.035 (0.024-0.046)	0 (0)	0	0.000	Reference
Weak	122.51 (72)	0.990	0.034 (0.024-0.045)	4.809 (4)	0	-0.001	Accept
Strong	128.53 (80)	0.991	0.032 (0.021-0.042)	6.02 (8)	0	-0.002	Accept
Strict	130.17 (84)	0.991	0.031 (0.02-0.04)	1.635 (4)	0	-0.002	Accept

Longitudinal invariance implies that changes in students' scores can be interpreted as true changes in their beliefs rather than artifacts of measurement. Table 4 reports the latent means from the strict model, with wave 1 means fixed to zero, so that the wave 2 estimates are the between-wave changes, tested with Wald tests. Latent means are in the metric of each factor's first indicator, and d expresses the change in wave 1 latent standard deviations. No dimension changed significantly between waves (all adjusted p > .30). Perceived meritocracy declined slightly (Δ = −0.29, p = .118, d = −0.14) and perceived privilege increased slightly (Δ = 0.15, p = .080, d = 0.13), a direction compatible with a growing perception of privilege, but neither difference is reliably different from zero. Preferences were practically unchanged, both for meritocracy (Δ = −0.08, p = .458) and for the acceptance of privilege (Δ = −0.12, p = .119). Overall, students' beliefs were stable over the period.

Table 4: Latent means of meritocracy and privilege dimensions by wave

Dimension	Wave 1	Wave 2 (SE)	z	p	p (Holm)	d
Perceived meritocracy (effort and talent)	0 (ref.)	-0.286 (0.183)	-1.56	.118	.354	-0.14
Perceived privilege (rich parents and contacts)	0 (ref.)	0.149 (0.085)	1.75	.080	.320	0.13
Preference for meritocracy (effort and talent)	0 (ref.)	-0.075 (0.101)	-0.74	.458	.458	-0.08
Acceptance of privilege (rich parents and contacts)	0 (ref.)	-0.115 (0.074)	-1.56	.119	.354	-0.10

Note: Latent means from the strict longitudinal invariance model (N = 591); wave 1 means are fixed to 0

(reference). Wave 2 estimates are the wave 2 − wave 1 differences, tested with Wald tests; p (Holm) adjusts for the four comparisons. d = difference divided by the latent standard deviation in wave 1.

4.4 Cohorts comparability

Table 5 reports the cross-cohort (multigroup) invariance tests comparing primary and secondary students, using wave 1 data. The configural model, which imposes the same four-factor structure in both cohorts, fits well, $\chi^2(28) = 47.65$, CFI = 0.991, RMSEA = 0.041. Weak invariance, however, was not achieved: constraining the factor loadings to be equal significantly worsened fit, $\Delta\chi^2(4) = 20.77$, $p < .001$, although the changes in CFI (ΔCFI = −0.009) and RMSEA (ΔRMSEA = 0.013) remain just within conventional cutoffs (Chen, 2007). Adding equality constraints on the thresholds, which play the role of intercepts for ordinal items, worsened fit further, $\Delta\chi^2(8) = 35.89$, $p < .001$, with ΔCFI = −0.011, which exceeds the recommended cutoff. The alternative sequence proposed by Svetina et al. (2020), which tests thresholds before loadings, led to the same conclusion (see supplementary material). The relationship between the latent factors and their indicators therefore differs between primary and secondary students, which could reflect developmental differences or differences in how the items are interpreted.

Table 5: Cohort (multigroup) invariance results

Model	χ^2 (df)	CFI	RMSEA (90% CI)	$\Delta\chi^2$ (Δdf)	ΔCFI	ΔRMSEA	Decision
Full invariance sequence							
Configural	47.65 (28)	0.991	0.041 (0.019-0.06)				Reference
Loadings	71.3 (32)	0.981	0.054 (0.037-0.071)	20.77 (4) ***	-0.009	0.013	Reject
Loadings + Thresholds	103.09 (40)	0.970	0.061 (0.047-0.076)	35.89 (8) ***	-0.011	0.007	Reject
Partial invariance (loading of preferences for talent freed)							
Loadings (partial)	48.48 (31)	0.992	0.037 (0.014-0.056)	1.82 (3)	0.001	-0.004	Accept
Loadings (partial) + Thresholds	84.91 (39)	0.978	0.053 (0.037-0.068)	42.72 (8) ***	-0.014	0.016	Reject

Note. Scaled WLSMV statistics. Each model is compared with the preceding one in its block (partial loadings: vs. configural).

To identify the source of non-invariance, we inspected univariate score tests for each equality constraint on the loadings. The misfit was concentrated in a single parameter: the loading of the talent preference item on the meritocratic preferences factor ($\chi^2(1) = 16.87$, Holm-adjusted $p < .001$). The remaining loadings were fully consistent with invariance (all $p > .36$). Accordingly, freeing that single loading is enough to restore weak invariance, with a fit indistinguishable from the configural model, $\Delta\chi^2(3) = 1.82$, $p = .61$, CFI = 0.992, RMSEA = 0.037.

The cohort-specific loadings (see supplementary material) show what this means

substantively. In primary school the meritocratic preferences factor is defined mainly by the preference for talent as a criterion of reward (standardized loading = 0.87) and much less by the preference for effort (0.45). In secondary school the pattern reverses: effort becomes the core of the factor (0.83) whereas talent is only weakly related to it (0.37). The meritocratic preferences factor thus does not carry an equivalent meaning across cohorts, which may reflect a change in how students interpret meritocratic principles over the school years, and its levels should be compared across cohorts with caution.

Even after freeing this loading, adding the threshold constraints significantly worsened fit ($\Delta\chi^2(8) = 42.72$, $p < .001$; $\Delta\text{CFI} = -0.014$), so strong invariance was also not achieved. In this case, the source is different: univariate score tests point to the thresholds of the perceived effort item, for which all three constraints are violated ($\chi^2(1)$ between 11.6 and 11.9; Holm-adjusted p between .015 and .017), whereas no threshold of any other item shows meaningful misfit (adjusted $p > .99$). Substantively, students in the two cohorts use the response categories of this item differently for a given level of perceived meritocracy, which is consistent with the differences reported in the descriptive section (see Figure 2): secondary students are more likely to endorse the lowest response category when judging whether effort is rewarded in society, suggesting that they apply different criteria. Taken together, only partial weak invariance holds, and comparisons of latent means across cohorts, in particular for meritocratic perceptions, should be interpreted with caution.

5 Discussion

Meritocracy and privilege are often treated as opposite poles of a single continuum, but the present study shows that they are better understood as distinct dimensions that coexist in students' minds, as found among adults (Castillo et al., 2023). As expected in H_1, the four latent constructs (perceived meritocracy, perceived privilege, meritocratic preferences, and privilege acceptance) are identifiable in the school-aged population, with satisfactory model fit and strong factor loadings in the two waves analyzed. Even at an early stage of school socialization, students distinguish between meritocratic and privilege-based perceptions and preferences. This structure resonates with qualitative evidence from Chilean schools, where students across socioeconomic groups simultaneously invoke meritocratic ideals and recognize structural advantage (Neut Aguayo et al., 2025), and where merit is often defined by attitude and visible effort rather than by academic outcomes (Moyano Dávila et al., 2025). Where merit is primarily understood as a behavioral disposition, accepting that privilege leads to better outcomes may not be experienced as a contradiction but as a separate normative judgment, which suggests that the coexistence of these orientations reflects how students navigate competing narratives about fairness and reward

The empirical separability of these dimensions, and the associations among them (H_2),

have implications for how meritocratic beliefs are understood and measured (Kwon & Pandian, 2024; Mijs, 2026; Zhu, 2025). First, perceived privilege is positively associated with stronger meritocratic preferences, consistent with a normative-reactive response to perceived unfairness (Castillo et al., 2023). Second, meritocratic perceptions are weakly correlated with privilege acceptance in the pooled sample, a pattern that is not consistent across cohorts and waves. Prior research has linked meritocratic perceptions to the self-interest hypothesis, whereby those in higher social positions are more likely to adopt a meritocratic worldview (Heuer et al., 2020). Incorporating privilege reveals a more complex picture: perceiving society as meritocratic may also conceal support for privilege, so that the justification of inequality is embedded in the meritocratic narrative itself (Khan & Jerolmack, 2013). These results are consistent with Zhu's (2025) argument that meritocratic and structural explanations accumulate rather than displace one another, with accounts in which merit endorsement remains stable as recognition of structural constraints increases (C. Liu & Wang, 2025; Tang et al., 2025).

These results also bear on how the literature on school meritocracy should be read. Research has linked stronger meritocratic beliefs among students to inequality legitimation, weaker support for redistribution, and greater acceptance of unequal access to educational resources (Batruch et al., 2022; Castillo et al., 2024; Wiederkehr et al., 2015), but mostly with unidimensional or undifferentiated measures that cannot establish whether these effects are specific to particular dimensions. Our findings suggest that they are likely dimension-specific: perceived meritocracy is not equivalent to preferred meritocracy, and neither carries the same consequences as perceived privilege or privilege acceptance. The present study provides a multidimensional measurement of meritocratic and privilege beliefs applicable to students, which makes it possible to examine whether distinct dimensions carry distinct consequences for the educational environment, from students' sense of justice in school to civic attitudes and the broader climate of fairness.

Regarding H_3, strict longitudinal invariance indicates that the scale measures the same constructs across waves, so that changes over time reflect real within-student change rather than shifts in how items are interpreted (Y. Liu et al., 2017; Putnick & Bornstein, 2016). Meritocratic belief structures appear fairly stable over one year in early-to-mid adolescence, which gives further studies in this agenda a sound measurement basis. Invariance across educational levels, however, was not supported. The non-invariance is concentrated in two parameters: the relative weight of talent and effort in meritocratic preferences, and how students read whether effort is rewarded. We do not consider this finding reducible to a mere measurement failure. Rather, we interpret it as older students attaching different meanings to these items, linked to their more sophisticated reasoning about fairness and to their growing exposure to school sorting such as grades, tracking, and peer comparison. The declining centrality of talent in older students' meritocratic

preferences may reflect a gradual disenchantment with inherited attributes (Resh & Sabbagh, 2014; Tang et al., 2025). This could be related to Chile's stratified system, where the gap between meritocratic discourse and structural reality can be experienced as a moral grievance (Neut Aguayo et al., 2025), but research in other contexts could help shed light on this issue.

6 Conclusions

Schools socialize students into beliefs about meritocracy, promoting effort and talent as the legitimate basis of reward while simultaneously exposing them to the weight of family resources and social connections in shaping outcomes (Goudeau & Croizet, 2017; Resh & Sabbagh, 2014; Traini, 2022). Many students navigate this tension by holding both orientations at once, endorsing merit as an ideal while recognizing that privilege structures actual outcomes, a configuration that can be associated with *dual consciousness* (Kluegel & Smith, 1987; C. Liu & Wang, 2025; Tang et al., 2025). Extending Castillo et al.'s (2023) multidimensional framework to two cohorts of Chilean students (6th and 9th graders in 2023, followed into 7th and 10th grade in 2024), the measurement of meritocratic and privilege beliefs provides evidence of reliability in school-aged populations, and equivalently over time. This is a foundational contribution: a precondition for studying the developmental trajectory of meritocratic beliefs and their educational implications, such as gender and social class differences.

This study draws on a non-probability sample of nine schools in one country, so its results cannot be statistically generalized, but three of its contributions may be relevant beyond this setting. First, the distinctions between perceived and preferred meritocracy, and between meritocratic and privilege-based principles, may reflect cognitive and normative capacities that develop during adolescence, although whether the four-factor structure replicates in other societies remains to be established. Second, the coexistence of merit endorsement and recognition of privilege has been documented in China (C. Liu & Wang, 2025; Tang et al., 2025), the United States and Finland (Zhu, 2025), and Europe (Kwon & Pandian, 2024), suggesting it is not a Chilean peculiarity. Third, longitudinal invariance testing as a prerequisite for developmental inference applies to any panel study of meritocratic beliefs. By contrast, the cross-cohort non-invariance may be more system-dependent, reflecting the intensity and timing of school-based sorting in Chile's tracked and marketized system (Madariaga, 2020; Valenzuela et al., 2013); research in countries with different tracking regimes will help clarify which aspects of measurement equivalence are universal.

This study has limitations that also point to directions for future research. First, the scale relies on two items per factor, which restricts both content coverage and reliability. Second, the design comprises only two waves and two school levels; additional waves

and a broader age range are needed to trace developmental trajectories and to establish when the meaning students attach to meritocracy begins to shift. Third, and most consequentially, the lack of invariance across school cohorts indicates that meritocracy scales cannot be assumed to function equivalently across educational stages. Future studies should therefore examine differential item functioning, develop stage-sensitive item formulations, and adopt partial-invariance approaches (Putnick & Bornstein, 2016; Van De Schoot et al., 2012). This would allow a more rigorous examination of group differences, such as the gender differences in meritocratic beliefs suggested by Gogescu (2025) and Khan & Jerolmack (2013), and of how beliefs about merit and privilege relate to the school outcomes for which they are likely to matter most: students' sense of justice, civic attitudes and academic motivation.

Data availability statement: The data and code that support the findings of this study are available in a GitHub repository. To preserve anonymity during peer review, the repository address is not included in this version of the manuscript; it can be provided to the editor upon request. Upon publication, the data and code will be deposited in a permanent public repository with a persistent identifier.

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